1. Introduction

The purpose of this practical assignment is to practice Linux command-line operations and understand how they are used in network administration. The assignment also involves developing and testing an AI Network Administrator Agent that executes approved network diagnostic commands and uses a local artificial intelligence model to analyze the collected results.

2. Objectives

The objectives of this assignment are:

- To practice basic Linux commands and navigate the file system.
- To create, copy, move, view, and remove files and directories.
- To use text-processing commands to search, sort, and analyze text.
- To inspect network interfaces, routing tables, and DNS configuration.
- To develop an AI agent that selects and executes an approved network diagnostic command.
- To understand the importance of validating AI-generated decisions before execution.

3. Linux Command-Line Practice

Linux commands are essential tools for system administration and network troubleshooting. The following commands were practiced in an Ubuntu virtual machine.

3.1 Basic Commands and Directory Navigation

Commands such as `pwd`, `ls`, `cd`, `date`, `whoami`, and `echo` help users identify their current location, list files, navigate directories, display the current date, identify the current user, and print text.

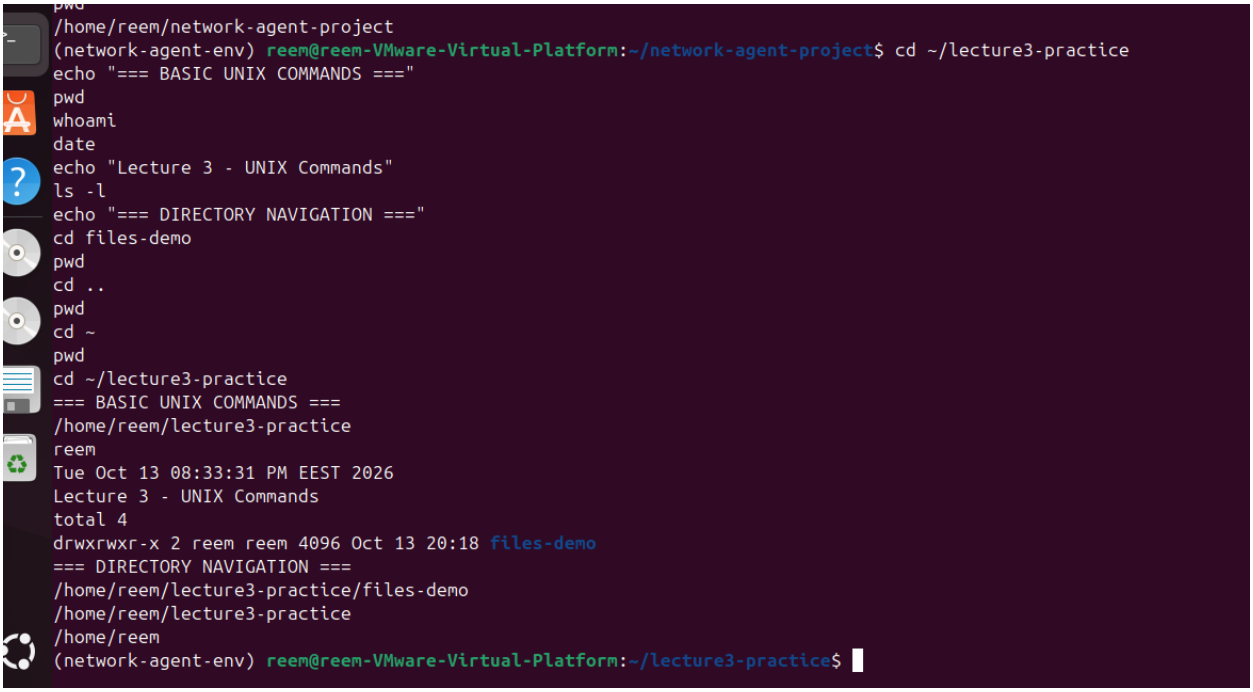


Figure 1: Basic Linux Commands

```bash

pwd

whoami

date

echo "Linux command practice"

ls -l

```

Explanation: These commands display basic system information and help users navigate and inspect the Linux environment.

3.2 Directory and File Management

Linux provides commands for creating directories, creating files, copying files, moving or renaming files, and deleting unnecessary files.

```bash
```

mkdir files-demo

echo "Linux file management practice" > file1.txt

cat file1.txt

```

*Explanation:* The `mkdir` command creates a directory, the `echo` command writes text into a file, and `cat` displays the file contents.

```bash

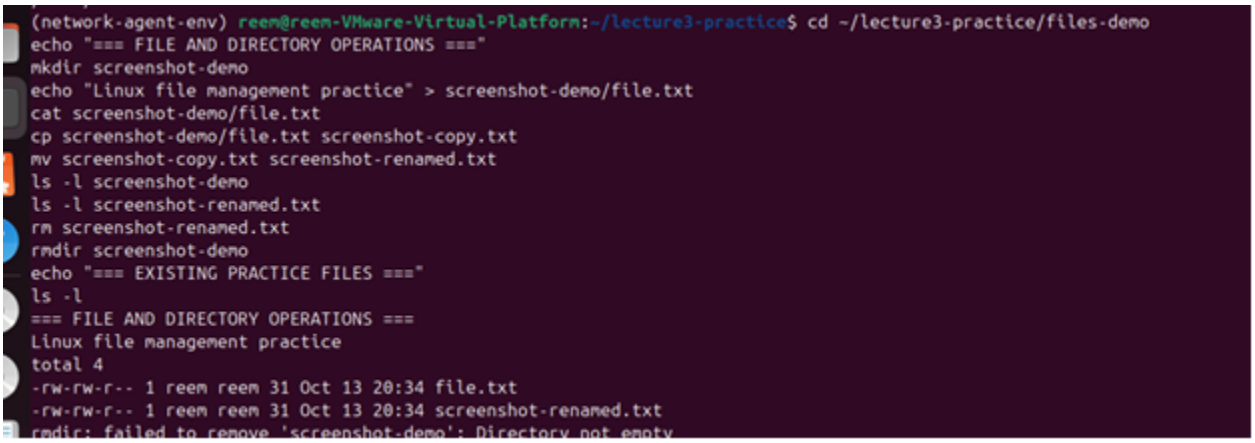
cp file1.txt copy.txt

mv copy.txt renamed.txt

ls -l

```

Explanation: The `cp` command copies a file, `mv` renames or moves a file, and `ls -l` displays detailed file information.




```bash

rm disposable.txt

mkdir empty-demo

rmdir empty-demo

```
```

Explanation: The `rm` command removes a file, while `rmdir` removes an empty directory. These commands should be used carefully to avoid deleting important data.

```
(network-agent-env) reem@reem-VMware-Virtual-Platform:~/network-agent-project$ cd ~/lecture3-practice/files-demo
(network-agent-env) reem@reem-VMware-Virtual-Platform:~/lecture3-practice/files-demo$ touch disposable.txt
mkdir empty-demo
ls -l disposable.txt empty-demo
-rw-rw-r-- 1 reem reem    0 Oct 13 21:57 disposable.txt

empty-demo:
total 0
(network-agent-env) reem@reem-VMware-Virtual-Platform:~/lecture3-practice/files-demo$ rm disposable.txt
rmdir empty-demo
(network-agent-env) reem@reem-VMware-Virtual-Platform:~/lecture3-practice/files-demo$ ls -ld disposable.txt empty-demo
ls: cannot access 'disposable.txt': No such file or directory
ls: cannot access 'empty-demo': No such file or directory
(network-agent-env) reem@reem-VMware-Virtual-Platform:~/lecture3-practice/files-demo$
```

```
(network-agent-env) reem@reem-VMware-Virtual-Platform:~/lecture3-practice/files-demo$ paste names.txt subjects.txt
ls -l
Alice   Cybersecurity
Bob     Networking
Reem    Linux
total 36
-rw-rw-r-- 1 reem reem   87 Oct 13 20:01 copy.txt
-rw-rw-r-- 1 reem reem   87 Oct 13 20:01 file1.txt
-rw-rw-r-- 1 reem reem   15 Oct 13 20:03 names.txt
-rw-rw-r-- 1 reem reem  145 Oct 13 20:03 notes.txt
-rw-rw-r-- 1 reem reem  118 Oct 13 20:05 reading.txt
-rw-rw-r-- 1 reem reem   37 Oct 13 20:01 renamed.txt
drwxrwxr-x 2 reem reem 4096 Oct 13 20:34 screenshot-demo
-rw-rw-r-- 1 reem reem   31 Oct 13 20:03 subjects.txt
-rw-rw-r-- 1 reem reem   54 Oct 13 20:20 vi-demo.txt
```

3.3 Viewing and Processing Text Files

Linux provides several commands for displaying file contents and processing text efficiently.

```
(network-agent-env) reem@reem-VMware-Virtual-Platform:~/lecture3-practice/files-demo$ cd ~/lecture3-practice/files-demo
echo "=== FILE VIEWING AND MERGING ==="
cat file1.txt
more reading.txt
less reading.txt
=== FILE VIEWING AND MERGING ===
Linux is useful for network administration.
Cybersecurity students use Linux commands.
Line 1: Linux basics
Line 2: Network administration
Line 3: Cybersecurity
Line 4: File systems
Line 5: Shell commands
(network-agent-env) reem@reem-VMware-Virtual-Platform:~/lecture3-practice/files-demo$ paste names.txt subjects.txt
ls -l
Alice   Cybersecurity
Bob     Networking
Reem    Linux
total 36
-rw-rw-r-- 1 reem reem   87 Oct 13 20:01 copy.txt
-rw-rw-r-- 1 reem reem   87 Oct 13 20:01 file1.txt
-rw-rw-r-- 1 reem reem   15 Oct 13 20:03 names.txt
-rw-rw-r-- 1 reem reem  145 Oct 13 20:03 notes.txt
-rw-rw-r-- 1 reem reem  118 Oct 13 20:05 reading.txt
-rw-rw-r-- 1 reem reem   37 Oct 13 20:01 renamed.txt
drwxrwxr-x 2 reem reem 4096 Oct 13 20:34 screenshot-demo
-rw-rw-r-- 1 reem reem   31 Oct 13 20:03 subjects.txt
-rw-rw-r-- 1 reem reem   54 Oct 13 20:20 vi-demo.txt
(network-agent-env) reem@reem-VMware-Virtual-Platform:~/lecture3-practice/files-demo$
```

Figure 5: Displaying File Contents

```bash
cat file1.txt
more reading.txt
less reading.txt
```

Explanation: The `cat` command displays file contents, while `more` and `less` allow users to view longer files page by page.

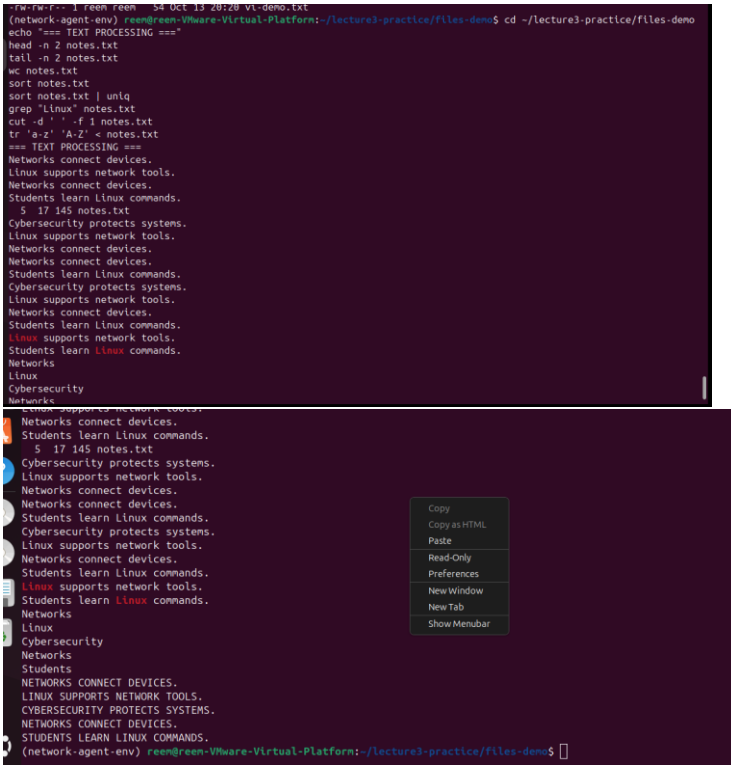
```bash
head -n 2 notes.txt
tail -n 2 notes.txt
wc notes.txt
```

Explanation: The `head` command displays the first lines of a file, `tail` displays the last lines, and `wc` counts lines, words, and bytes.

```bash
sort notes.txt
sort notes.txt | uniq
```

Explanation: The `sort` command arranges text lines alphabetically. The `uniq` command removes adjacent duplicate lines, so sorting first helps identify repeated lines throughout a file.

```bash
grep "Linux" notes.txt
cut -d ' ' -f 1 notes.txt
tr 'a-z' 'A-Z' < notes.txt
```



Explanation: The `grep` command searches for matching text, `cut` extracts selected fields, and `tr` transforms characters, such as converting lowercase letters to uppercase.

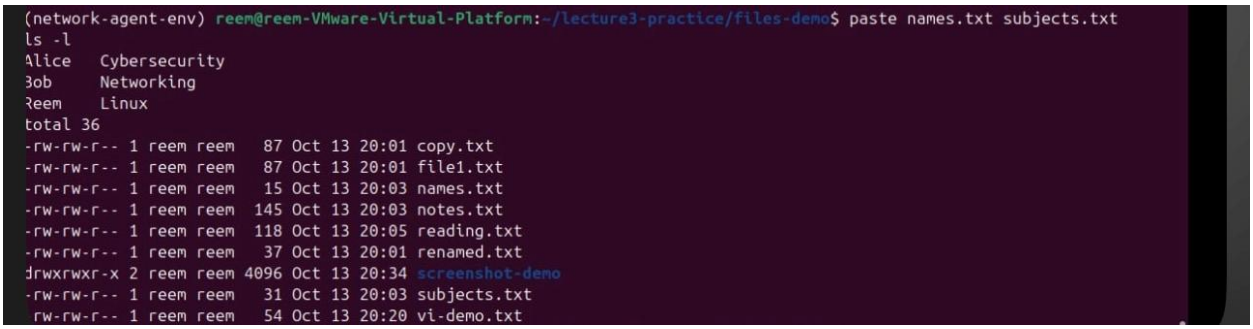


Figure 9: Combining Text from Multiple Files

```bash
paste names.txt subjects.txt
```

Explanation: The `paste` command combines corresponding lines from two files side by side. In this exercise, names were displayed alongside their associated subjects.

4. AI Network Administrator Agent

4.1 Description and Implementation

An AI Network Administrator Agent was implemented in Python and tested on Ubuntu. The program uses Python's `subprocess` module to execute approved diagnostic commands and `urllib.request` to communicate with a locally running Ollama service using the `qwen2.5:3b` model.

The original agent collects real network information, including interface addresses, routing information, DNS configuration, gateway connectivity, and domain-name resolution. The collected outputs are then provided to the AI model to generate a diagnostic report.

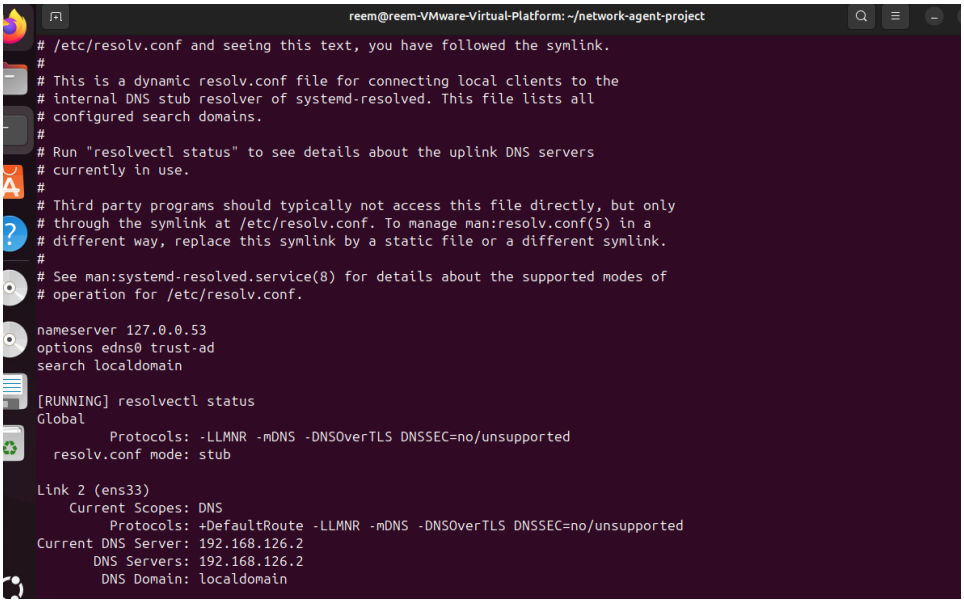
```bash
cd ~/network-agent-project
python network_agent.py
```

```
(network-agent-env) reem@reem-VMware-Virtual-Platform:~/lecture3-practice/files-demo$ cd ~/network-agent-project
python network_agent.py
============================================================
AI NETWORK ADMINISTRATOR AGENT
============================================================

[RUNNING] ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:28:64:e6 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.126.128/24 brd 192.168.126.255 scope global dynamic noprefixroute ens33
       valid_lft 1647sec preferred_lft 1647sec
    inet6 fe80::ff95:6bed:c299:5711/64 scope link noprefixroute
       valid_lft forever preferred_lft forever

[RUNNING] ip route
default via 192.168.126.2 dev ens33 proto dhcp src 192.168.126.128 metric 100
192.168.126.0/24 dev ens33 proto kernel scope link src 192.168.126.128 metric 100

[RUNNING] cat /etc/resolv.conf
# This is /run/systemd/resolve/stub-resolv.conf managed by man:systemd-resolved(8).
# Do not edit.
#
# This file might be symlinked as /etc/resolv.conf. If you're looking at
```



Explanation: This command runs the original agent. The program executes predefined diagnostic commands, displays their outputs, and sends the collected information to the AI model for analysis.

4.2 Network Diagnostic Commands

The original agent uses the following commands:

| Command | Purpose |

| `ip addr` | Displays network interfaces and IP addresses. |

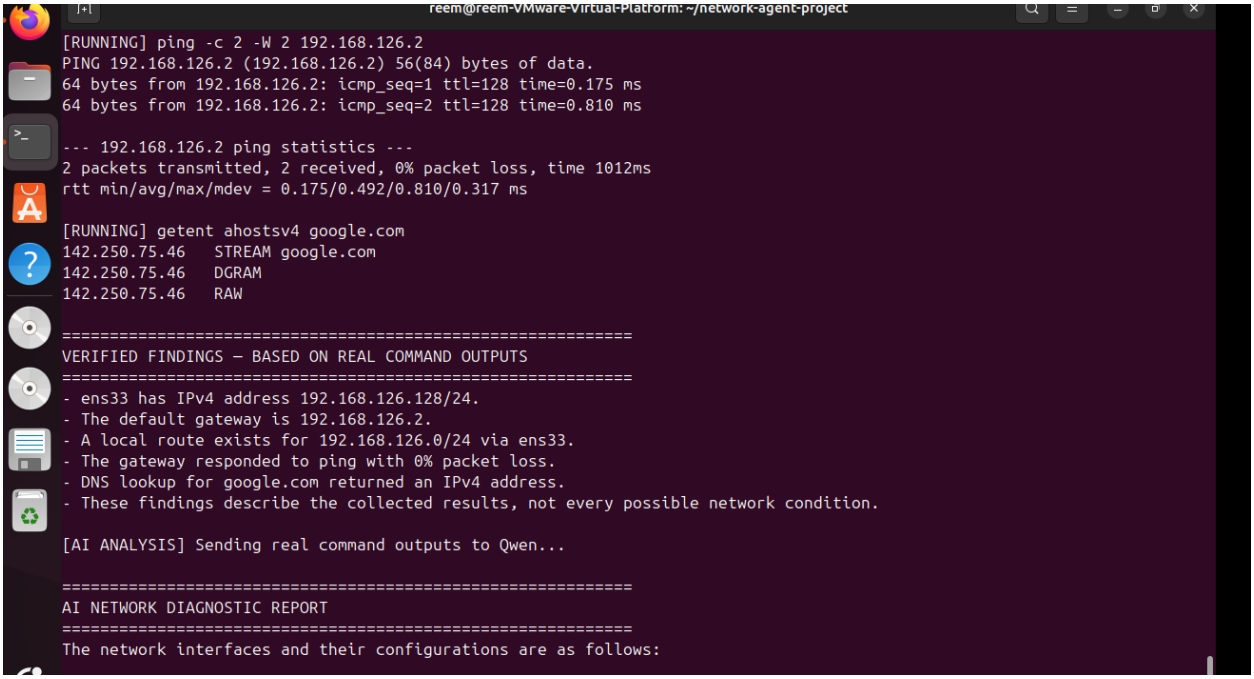
| `ip route` | Displays the routing table and default gateway. |

| `cat /etc/resolv.conf` | Displays the local DNS resolver configuration. |

| `resolvectl status` | Displays the active DNS configuration. |

| `ping -c 2 -W 2 192.168.126.2` | Tests connectivity to the default gateway. |

| `getent ahostsv4 google.com` | Checks whether an IPv4 address can be resolved for a domain name. |

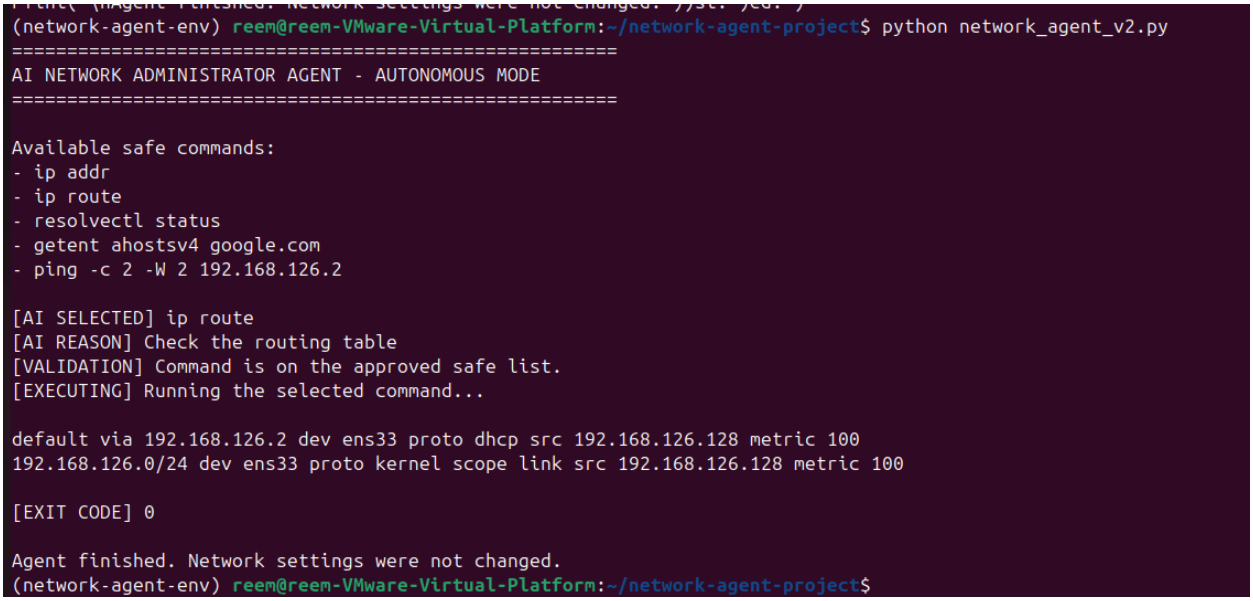


Explanation: The agent identified the network interface `ens33` with IPv4 address `192.168.126.128/24` and default gateway `192.168.126.2`. The gateway responded successfully to the ping test, with 0% packet loss. The domain lookup also returned an IPv4 address for `google.com`.

The results indicate that the tested gateway connectivity and domain-name resolution were working during the experiment.

4.3 AI-Based Command Selection and Execution

A second version of the program, `network_agent_v2.py`, was created to demonstrate a more autonomous workflow. The AI model selects one command from a predefined list of permitted diagnostic commands. The Python program validates the selected command against the approved list before executing it.




```bash
python network_agent_v2.py
```

Explanation: During the test, the AI model selected the `ip route` command to inspect the routing table. The program verified that the command was permitted, executed it, and displayed the routing information.

The output confirmed the default route through `192.168.126.2` and the local network route `192.168.126.0/24` through `ens33`. The command completed with exit code `0`, indicating successful execution.

The agent uses an allowlist of approved read-only commands. This approach limits the commands that the AI can execute and helps prevent unintended changes to the system. The agent reported that no network settings were changed.

5. Results and Discussion

The practical exercises demonstrated how Linux commands support file management, text processing, and network troubleshooting. The network diagnostic agent successfully collected real command outputs and used them to generate an AI-based analysis.

The autonomous version demonstrated that an AI model can select an appropriate diagnostic command from an approved list and execute it through a Python program. However, the model is not permitted to execute arbitrary commands. This restriction is an important safety measure when developing AI agents for system administration.

The experiment also demonstrated that AI-generated analysis should be checked against actual command outputs because AI models may occasionally provide incomplete or inaccurate interpretations.

6. Conclusion

This assignment provided practical experience with Linux command-line tools and network administration tasks. It also demonstrated how Python and a locally running AI model can be combined to build a network diagnostic agent.

The agent successfully inspected network configuration, tested gateway connectivity, checked domain-name resolution, and selected and executed an approved diagnostic command. Overall, the assignment improved understanding of Linux administration, network troubleshooting, AI-assisted automation, and the importance of restricting automated actions to safe and validated commands.

Reem saifi

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